

CASE STUDY

HR Analytics Performance Dashboard

Turning Workforce Data into Retention Strategy Using Microsoft Excel

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Tools: Microsoft Excel · Pivot Tables · Pivot Charts · Slicers · VBA

Category: HR Analytics / Business Intelligence / Workforce Analytics

1. Project Overview

This case study documents the design and delivery of an HR Analytics Performance Dashboard built in Microsoft Excel for a mid-sized organization's People & Culture function. The project converts a raw, 171-record employee dataset into a single interactive dashboard that reports on headcount, attrition, internal mobility, diversity, job-level distribution, employment type, manager span of control, and workforce tenure — all filterable in real time through slicers.

The engagement was structured as a full analytics lifecycle: source-data audit, cleaning and standardization, pivot-based data modeling, KPI definition, dashboard design, and stakeholder-ready delivery, including a presentation mode and a one-click VBA refresh. The deliverable was designed to be used in monthly HR leadership reviews, replacing what had previously been a set of disconnected manual spreadsheet exports.

At a glance

Item	Detail
Client function	HR / People Analytics
Data volume	171 employee records, 16 fields, 2020–2025
Primary tool	Microsoft Excel (Pivot Tables, Pivot Charts, Slicers, VBA)
Delivery format	Single-page interactive executive dashboard
Refresh method	One-click VBA macro across 8 linked pivot tables
Core KPIs	Overall HC, Active HC, Attrition %, Transfer, Diversity %, Job Level %, Stability %

2. Business Background

The organization had grown from 23 active employees in 2020 to 118 by 2025, a more than five-fold increase in headcount over five years. Growth of this pace typically outstrips the reporting infrastructure supporting it, and this organization was no exception: HR headcount and attrition figures were being manually compiled per department, per request, with no standard, shared source of truth. Different stakeholders frequently arrived at leadership meetings with slightly different numbers for the same reporting period, undermining confidence in the data.

At the same time, the business had never formally segmented its attrition analysis below the company-wide level. A single, blended attrition percentage was reported quarterly, with no visibility into which departments, managers, or employment types were actually driving it. This created a structural blind spot: leadership could see that turnover existed, but had no reliable way to target retention investment where it would have the greatest impact.

3. Business Problem

HR leadership needed a single, trustworthy, and interactive view of workforce health that could answer four recurring questions without a manual data pull each time:

- How many people does the organization actually employ right now, versus how many have ever been employed?
- Is attrition evenly distributed across the business, or concentrated in specific departments, managers, or employment types?
- Is the organization retaining talent through internal mobility, or losing it entirely through external attrition?
- Is the workforce's seniority and tenure structure healthy, or top-light and short-tenured in a way that signals future risk?

Framed as a single problem statement: the business lacked a reliable, drillable view of workforce composition and attrition that could distinguish a genuine organization-wide retention problem from a small number of concentrated, addressable hotspots.

4. Project Objectives

- **Establish one source of truth.** Build a single clean data model that every KPI, chart, and manager review draws from, eliminating conflicting manual reports.
- **Quantify attrition precisely.** Report both a company-wide attrition rate and the department-, manager-, and employment-type-level rates hiding beneath it.
- **Make the dashboard interactive, not static.** Allow any HR business partner to filter by Name, Department, Reporting Manager, or Year without editing a single formula.
- **Support two audiences with one file.** Design a working/analysis view for HR analysts and a clean presentation view for leadership readouts.
- **Remove manual maintenance overhead.** Automate the refresh of all pivot tables and charts with a single VBA-triggered action.
- **Produce an auditable, explainable model.** Ensure every KPI's calculation logic is transparent and defensible under scrutiny from Finance or executive stakeholders.

5. Dataset Description

The dataset consists of 171 employee-level records spanning hire years 2020 through 2025, stored as a single structured Excel Table (Table2) with 16 fields.

Field	Description
Emp Status	On-Board / Attrition / Transfer — the master status flag every KPI filters on
Movement	Internal Job Posting (IJP) or OnRoll flag identifying internal moves
Emp Id / Name	Unique employee identifier and display name

Field	Description
Date of Joining / Date of Relieving	Hire date and exit date (exit date populated only for Attrition records)
Reporting Manager	Current or last manager of record
Emp Type	Permanent or Contract
Category (Gender)	Male / Female
Date of Birth	Used for age-related segmentation
Marital Status	Single / Married
Grade	Job level: Contract, Trainee, Professional, Management
Department	Accounting, Finance, HR, Legal, Marketing, Operations, Sales
Total Experience	Prior work experience in years at time of hire
Year	Formula-derived year of joining
Stability	Formula-derived tenure bucket (<1, 1+, 2+, 3+, 4+, 5+ years)

Overall Headcount across the full dataset is 171; Active Headcount (Emp Status = On-Board) is 118; 47 records are Attrition; 6 are Transfer.

6. Data Cleaning Process

Before any pivot or chart was built, the raw export was audited and standardized. This step consumed a larger share of project time than the dashboard build itself — consistent with typical analytics engagements, where data preparation routinely accounts for the majority of total effort.

- **Status standardization:** Confirmed every record carried exactly one of three valid Emp Status values (On-Board, Attrition, Transfer), since this single field is the master filter for every downstream KPI — any inconsistency here would silently corrupt every pivot in the workbook.
- **Date field integrity:** Verified Date of Joining was populated for all 171 records and that Date of Relieving was populated only where Emp Status = Attrition, catching any record where an exit date existed without a corresponding status change.
- **Categorical field normalization:** Standardized free-text-prone fields (Department, Grade, Emp Type, Category) to a fixed, consistent set of values so pivot groupings would not fragment into near-duplicate categories (e.g., "HR" vs. "Human Resources").
- **Header hygiene review:** Flagged a header typo ("Data of Relieving") that, while cosmetic, poses a real risk to any future formula or Power Query step that references columns by name rather than position.
- **Duplicate and orphan-record check:** Confirmed Emp Id uniqueness across all 171 rows and checked that every Reporting Manager value referenced a plausible, real manager rather than a data-entry variant.
- **Structured table conversion:** Converted the raw range into a proper Excel Table (Table2) rather than a static range, so that new rows added in future refresh cycles are automatically included in every pivot without manual range edits.

7. Data Transformation

Two fields were engineered rather than manually entered, both designed to keep the dashboard self-maintaining as time passes and the workbook is reopened in future periods.

- **Year (hire year):** Derived using `=YEAR(OFFSET(Table2,row,4,1,1))` directly from Date of Joining, guaranteeing the hire-year grouping used in the Head Count Growth chart always matches the actual join date, even if that date is later corrected.
- **Stability (tenure bucket):** Derived using a nested `IF(DATEDIF(DateOfJoining, TODAY(), "Y") >= N, ...)` formula ladder, automatically classifying every employee into a <1, 1+, 2+, 3+, 4+, or 5+ year tenure band that recalculates every time the workbook is opened.

Beyond field-level transformation, the core structural transformation was building eight interconnected Pivot Tables — Year × Status, Grade, Stability, Reporting Manager, Emp Type, Department, Gender, and a manually assembled KPI/percentage summary table — each filtered appropriately (active-only for workforce-composition views; full historical population for flow and tenure views), so that every chart on the dashboard reads from a governed, consistent model rather than an ad hoc formula.

8. Dashboard Design Decisions

- **Single-page executive layout:** All KPIs and charts were fitted onto one screen to support a genuine at-a-glance leadership review, avoiding the need to scroll or tab between sheets during a live meeting.
 - **KPI cards above chart detail:** The five headline numbers (Overall HC, Active HC, Attrition, Attrition %, Transfer) were placed in a fixed strip at the top, so viewers anchor on the topline figures before drilling into composition charts below.
 - **Consistent dark theme with accent colors:** A dark background with blue, orange, and green accents was used throughout to keep every chart visually consistent and to make status-based color coding (e.g., Attrition vs. On-Board vs. Transfer) instantly recognizable across every chart on the page.
 - **Chart-type selection matched to data shape:** Ranked comparisons (Head Count Reporting) used horizontal bar charts; ordered categorical data (Stability, Job Level) used column charts; simple two-category splits (Diversity, Employee Type) used donut/stacked formats.
 - **Conditional formatting for emphasis:** Applied to KPI values and reporting tables to visually flag values outside expected ranges without requiring a separate written commentary.
 - **Slicer placement for guided exploration:** Slicers were grouped on the left rail in a fixed order (Name, Reporting Manager, Year, Department) reflecting the most common analysis path — from a specific person, to their manager, to a time period, to a department — rather than an arbitrary order.
 - **Presentation Mode:** A secondary, simplified view was built specifically for stakeholder walkthroughs, hiding working cells and pivot mechanics so the delivered experience looks like a finished product rather than a spreadsheet.
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9. KPI Definitions

KPI	Definition & Formula
Overall HC	COUNT of all employee records regardless of status = 171
Active HC	COUNTIFS(Emp Status = "On-Board") = 118
Attrition (count)	COUNTIFS(Emp Status = "Attrition") = 47
Attrition %	Attrition ÷ Overall HC = 47 ÷ 171 = 27.5% (lifetime-cumulative rate, not annualized)
Transfer	COUNTIFS(Emp Status = "Transfer") = 6
Diversity %	COUNTIFS(Category, Emp Status="On-Board") ÷ Active HC, by gender
Job Level %	COUNTIFS(Grade, Emp Status="On-Board") ÷ Active HC, by grade
Employee Type %	COUNTIFS(Emp Type, Emp Status="On-Board") ÷ Active HC
Department HC %	COUNTIFS(Department, Emp Status="On-Board") ÷ Active HC
Head Count Reporting	COUNTIFS(Reporting Manager, Emp Status="On-Board"), by manager
Stability %	COUNT of tenure bucket (DATEDIF-based) ÷ Overall HC, across the full historical population

Each KPI's filter basis is deliberate: workforce-composition KPIs (Diversity, Job Level, Employee Type, Department, Reporting) describe the current organization and are therefore filtered to active employees only, while flow and tenure KPIs (Attrition, Transfer, Stability) describe the organization's full history and are intentionally left unfiltered by status.

10. Business Insights

Segmenting the headline 27.5% attrition rate by department, manager, and employment type surfaced patterns that were completely invisible in the company-wide figure alone.

Segment	Attrition Rate	Business Meaning
Accounting (department)	50.0%	Nearly double the company average — the single largest concentration of turnover risk in the business
HR (department)	32.4%	Above average; the function responsible for retention strategy is itself experiencing elevated turnover
Finance (department)	28.6%	Slightly above the company average
Operations (department)	29.8%	Above average, and the organization's largest department by headcount
Marketing (department)	21.1%	Below average
Legal (department)	15.4%	Well below average

Segment	Attrition Rate	Business Meaning
Sales (department)	8.7%	The most stable department in the business
Contract (employment type)	43.5%	Nearly double the Permanent rate
Permanent (employment type)	25.0%	Close to the company average
Female (gender)	29.7%	Modestly higher than male attrition
Male (gender)	26.2%	Modestly lower than female attrition

Manager-level analysis reinforced this pattern: three managers (Alejandra Mack, Janelle Wiley, and Jefferson Perry) each carry team attrition rates above 36%, materially higher than the company average, while the manager with the largest team by far (Ryan Simmons, 42 direct/indirect reports) has the lowest attrition rate of any manager at 19.0% — demonstrating that span of control alone does not predict retention outcomes in this organization; manager- and team-specific factors are the stronger driver.

11. Key Findings

- **Finding 1 — Attrition is concentrated, not distributed.** A department-level attrition rate nearly double the company average (Accounting, 50.0%) was hidden inside a moderate-looking blended figure of 27.5%.
- **Finding 2 — Contractors are a disproportionate churn risk.** At 43.5% attrition versus 25.0% for permanent staff, the contract workforce — while only 11% of active headcount — is nearly twice as likely to exit.
- **Finding 3 — Manager quality outweighs team size as a retention factor.** The organization's widest-span manager has its lowest attrition rate, while three narrower-span managers have the highest, indicating team-specific management practices matter more than raw span of control.
- **Finding 4 — Internal mobility is underused.** Only 6 transfers were recorded against 47 exits, meaning the organization is losing talent externally at nearly eight times the rate it is redeploying talent internally.
- **Finding 5 — The workforce pyramid is bottom-heavy.** 46% of active employees sit in the Trainee or Contract tier, the least senior and least secure segment, signaling future promotion-pipeline pressure if not addressed.
- **Finding 6 — Gender representation is imbalanced with an emerging attrition gap.** A 62/38 male-to-female split is compounded by a modest but consistent gap in female attrition (29.7% vs. 26.2%).
- **Finding 7 — Long-tenured retention is genuinely strong where it exists.** 32% of everyone the organization has ever employed reached the five-year tenure mark, showing the organization can retain talent well under the right conditions — the problem is inconsistency, not an inability to retain.

12. Business Recommendations

- **1. Launch a targeted Accounting retention review.** Conduct structured exit interviews for recent Accounting leavers and benchmark compensation and workload against Sales, the most stable department, to isolate the root cause.

- **2. Reassess the contractor engagement model.** Determine what share of contractor exits are voluntary versus planned contract-end, and pilot a conversion-to-permanent pathway for high-performing contractors.
- **3. Introduce manager-level retention accountability.** Add team attrition rate as a formal input to the performance reviews of the three highest-attrition managers, paired with structured HRBP coaching support.
- **4. Formalize an internal mobility program.** Launch an internal job board with a policy that removes the need for a current manager's approval before an employee can apply internally, directly addressing the 6-to-47 transfer-to-attrition imbalance.
- **5. Build a defined promotion pathway from Trainee to Professional.** Establish clear timelines and criteria to relieve pressure on the bottom-heavy job-level pyramid before it compounds into further attrition.
- **6. Audit sourcing and job-description language for gender bias.** Particularly in Operations and Sales-heavy hiring pipelines, to begin narrowing the 62/38 gender split over time.
- **7. Reframe attrition reporting at the leadership level.** Present Attrition % alongside its department- and employment-type-level breakdown by default, rather than as a single blended company-wide figure, to prevent under-reaction to concentrated hotspots.

13. Challenges Faced

Several challenges emerged across the data, design, and stakeholder dimensions of the project, each requiring a distinct resolution approach.

- **Inconsistent source data:** Status labels, department names, and date fields required standardization before any pivot could be trusted.
- **Conflicting stakeholder expectations:** HR wanted department-level detail; leadership wanted a single clean summary. The dashboard needed to serve both audiences without becoming cluttered.
- **A misleading headline metric:** The blended attrition rate looked acceptable on its own and risked leadership complacency unless the dashboard was explicitly designed to surface the concentration underneath it.
- **Balancing depth with simplicity:** Enough analytical depth was needed to support real HR decisions, without overwhelming a non-technical executive audience during a live walkthrough.
- **Maintaining a single source of truth:** With eight separate pivot tables in play, ensuring every one used a consistent filter logic (active vs. full-population) required careful, explicit documentation to avoid silent inconsistencies.

14. Technical Challenges

- **Slicer connectivity across multiple pivots:** Slicers do not automatically filter every pivot table built from the same source; each slicer had to be explicitly linked to all eight pivots via Report Connections to ensure the dashboard behaved as a single, unified interface.

- **Formula-driven fields recalculating on file open:** The Stability field's use of TODAY() means tenure buckets shift every time the workbook is reopened, which had to be explicitly documented so viewers understood the dashboard reflects a live, not fixed, snapshot.
- **Chart-type distortion risk:** Early drafts used 3D column and pie charts for visual appeal; these were reviewed against standard data-visualization guidance, since 3D perspective is known to distort perceived value at a glance, and retained only where the category count was low enough (two-way splits) for the distortion risk to be minimal.
- **Pivot cache duplication and performance drag:** As additional pivot tables were added, redundant pivot caches began slowing file performance; caches were consolidated and unnecessary conditional formatting rules removed to restore responsiveness.
- **Structured table range management:** Ensuring the source range remained a true Excel Table (rather than a static range) was necessary so future data additions would be picked up automatically by every pivot without manual range redefinition.

15. VBA Automation

A single VBA macro was built to remove the manual burden of refreshing eight separate pivot tables individually every time the underlying data changed.

- **Refresh All macro:** Iterates through every PivotCache in the workbook and refreshes it in one action, ensuring every chart and KPI updates in sync rather than risking a partially refreshed, internally inconsistent dashboard.
- **Presentation Mode toggle:** A macro-driven view switch hides working/analysis elements and activates a clean, stakeholder-facing layout for live reviews, without requiring a second physical copy of the file to be maintained.
- **Why VBA rather than a manual process:** With eight interconnected pivots, a manual refresh is easy to execute incompletely (e.g., refreshing six of eight pivots and presenting outdated numbers on the remaining two); a single macro-triggered refresh removes that risk entirely and reduces refresh time from several manual steps to one click.

17. Macros VBA Code {Full Screen}

Option Explicit

Public DashboardMode As Boolean

Sub ToggleDashboardMode()

 If DashboardMode = False Then

 With Application

 .DisplayFormulaBar = False

 .DisplayStatusBar = False

 .ExecuteExcel4Macro "SHOW.TOOLBAR(\"\"Ribbon\"\",False)"

 End With

With ActiveWindow

.DisplayWorkbookTabs = False

.DisplayGridlines = False

.DisplayHeadings = False

.DisplayHorizontalScrollBar = False

.DisplayVerticalScrollBar = False

End With

FitDashboard

DashboardMode = True

Else

With Application

.DisplayFormulaBar = True

.DisplayStatusBar = True

.ExecuteExcel4Macro "SHOW.TOOLBAR(\"\"Ribbon\"\",True)"

End With

With ActiveWindow

.DisplayWorkbookTabs = True

.DisplayGridlines = True

.DisplayHeadings = True

.DisplayHorizontalScrollBar = True

.DisplayVerticalScrollBar = True

.Zoom = 100

End With

DashboardMode = False

End If

End Sub

Sub FitDashboard()

Dim DashRange As Range

Dim DashWidth As Double

Dim WindowWidth As Double

Dim Z As Long

Set DashRange = Worksheets("HR Dashboard").Range("A1:AA42")

DashWidth = DashRange.Width

WindowWidth = ActiveWindow.UsableWidth

Z = Int((WindowWidth / DashWidth) * 100)

If Z > 130 Then Z = 130

If Z < 60 Then Z = 60

```

ActiveWindow.Zoom = Z
ActiveWindow.ScrollRow = 1
ActiveWindow.ScrollColumn = 1
Worksheets("HR Dashboard").Range("A1").Select
End Sub

```

16. Lessons Learned

- **Data cleaning is the majority of the work, not a preliminary step.** The time spent standardizing and validating the source data exceeded the time spent building the visual layer, which reframed how project time should be planned in future work.
- **A defensible KPI is more valuable than an impressive one.** The most important design decision in the project was not a chart choice but the decision to disclose exactly how Attrition % is calculated, since an undocumented denominator is a credibility risk under executive scrutiny.
- **The average is rarely where the real story lives.** Every meaningful finding in this project came from segmenting a topline number, not from the topline number itself — a principle that will shape how future dashboards are structured from the outset, not added as an afterthought.
- **Design choices have measurement consequences.** Chart type is not a purely aesthetic decision; 3D effects and angle-based encodings (pies, donuts) carry real perceptual accuracy trade-offs that must be weighed against visual appeal.
- **Automation pays for itself quickly.** A relatively small VBA investment (a single refresh macro) removed a recurring, error-prone manual task for every future reporting cycle.

17. Future Enhancements

- **Power BI migration:** Rebuild the model as a proper star schema with DAX measures, enabling native drill-through pages, bookmarks, and more flexible time-intelligence calculations than Excel pivots support.
- **SQL integration:** Connect the model to a live SQL Server or PostgreSQL backend rather than a static file, enabling scheduled, automated data refresh from the source HRIS.
- **Python-based automation:** Use pandas for data cleaning and validation steps currently performed manually, improving repeatability and auditability.
- **Predictive attrition analytics:** Build a classification model scoring each active employee's exit probability using tenure, department, manager, and job level as features, moving the dashboard from descriptive to predictive.
- **Employee churn prediction:** Extend the predictive model into a standing churn-risk score refreshed on each data cycle, flagging at-risk employees before they resign rather than reporting on attrition after the fact.
- **HR forecasting:** Apply time-series methods to the existing Year × Status pivot to project next year's expected attrition and corresponding hiring requirements.

- **Executive reporting layer:** Add an auto-generated plain-language summary translating each period's KPI movement into leadership-ready bullet points.
- **Real-time dashboard:** Move from a periodically refreshed static file to a continuously live-connected reporting layer.
- **Cloud data source:** Host the underlying data in a cloud environment (e.g., Azure or a managed data warehouse) to support secure, multi-user, anywhere access.
- **Expanded data capture:** Add compensation band, performance rating, and engagement/eNPS score to the data model to enable pay-equity, performance-correlation, and satisfaction-driven analysis not currently possible with headcount data alone.

18. Portfolio Summary

This project demonstrates the complete analytics lifecycle applied to a real business function: auditing and cleaning inconsistent source data, building a governed multi-pivot data model, defining transparent and defensible KPIs, designing an interactive dual-audience dashboard, and automating maintenance through VBA. More importantly, it demonstrates the analytical instinct to look past a single headline metric — the project's central finding, a department-level attrition rate nearly double the company average hidden inside a moderate blended figure, was only discoverable through deliberate segmentation, not visible in the topline number at all.

The deliverable reflects both technical craftsmanship (pivot architecture, slicer connectivity, formula-driven dynamic fields, VBA automation) and business fluency (translating a data pattern into a specific, actionable HR recommendation rather than stopping at description). It was built as a foundation for further growth into Power BI, SQL, and predictive HR analytics, and stands as evidence of the ability to deliver an end-to-end analytics product, not just a chart.

Tools demonstrated: Microsoft Excel · Pivot Tables · Pivot Charts · Slicers · Conditional Formatting · Dashboard Design · VBA Macros · Data Cleaning · Data Analysis · Business Intelligence